

Genome-wide fusion-hotspot cohort scan: msk_impact_50k_2026

544 of 3919 genes passed the ≥ 5 -patient recurrence gate; 4 FDR-significant ($q < 0.05$).

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Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
NTRK3	curated	ok	71	98.21	96.49	9.002e-09	yes	0.4884
ROS1	curated	ok	204	72.73	90.68	0.000244	yes	0.4271
ETV6	curated	ok	213	81.01	77.27	0.002572	yes	0.4489
FLI1	curated	ok	121	68.22	94.02	0.009047	yes	0.5858
RET	curated	ok	219	85.38	92.27	0.1066	no	0.4454
FGFR2	curated	ok	188	93.1	85.82	0.1066	no	0.3353
TMPRSS2	auto	ok	1088	33.33	6.243	0.1675	no	0.5306
ERG	curated	ok	863	32.96	97.68	0.1675	no	0.5179
EWSR1	auto	ok	415	78.35	11.3	0.1675	no	0.2932
ALK	curated	ok	321	97.41	96.31	0.1675	no	0.6192
TP53	auto	ok	258	30.77	27.91	0.1675	no	0.235
BRAF	curated	ok	247	92.64	91.57	0.1675	no	0.3296
EML4	auto	ok	224	97.51	53.36	0.1675	no	0.6667
FGFR3	auto	ok	194	97.37	94.04	0.1675	no	0.4923
TACC3	auto	ok	138	97.06	48.48	0.1675	no	0.5678
NOTCH1	auto	ok	115	57.89	60.87	0.1675	no	0.3243
NTRK1	curated	ok	113	72.73	84.21	0.1675	no	0.3578
NAB2	auto	ok	111	71.43	69.01	0.1675	no	0.3891
CDK12	auto	ok	99	50	39.02	0.1675	no	0.2757
NOTCH2	auto	ok	92	25	53.85	0.1675	no	0.2276
TERT	auto	ok	92	0	81.25	0.1675	no	0.3993
ERBB2	auto	ok	88	35	43.18	0.1675	no	0.2224
MET	auto	ok	73	89.47	75	0.1675	no	0.3205
POLE	auto	ok	73	36.36	31.82	0.1675	no	0.2072
FGFR1	auto	ok	64	58.33	50	0.1675	no	0.2314
STAT6	auto	ok	60	100	6	0.1675	no	0.3594
DNAJB1	auto	ok	54	100	2.564	0.1675	no	0.5432
RNF43	auto	ok	48	60	38.89	0.1675	no	0.2448
KIAA1549	auto	ok	44	97.56	50	0.1675	no	0.5302
POLD1	auto	ok	36	33.33	46.15	0.1675	no	0.2352

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
RBM10	auto	ok	36	75	71.43	0.1675	no	0.347
PRKACA	auto	ok	35	96	96.67	0.1675	no	0.6812
DNMT3A	auto	ok	32	100	36.36	0.1675	no	0.3288
ATF1	auto	ok	30	84	90	0.1675	no	0.4076
BICC1	auto	ok	29	100	67.86	0.1675	no	0.4255
STAT5B	auto	ok	27	66.67	75	0.1675	no	0.2326
RPS6KA4	auto	ok	21	62.5	58.33	0.1675	no	0.2143
ASPCR1	auto	ok	20	93.33	20	0.1675	no	0.4833
LMNA	auto	ok	17	100	38.46	0.1675	no	0.4387
AGK	auto	ok	14	100	14.29	0.1675	no	0.622
TPM3	auto	ok	14	100	64.29	0.1675	no	0.5362
NR4A3	auto	ok	7	100	71.43	0.1675	no	0.3723
EMID1	auto	ok	5	75	25	0.1675	no	0.4674
EGFR	auto	ok	466	70	74.55	0.1917	no	0.3306
EZH1	auto	ok	30	0	20	0.2056	no	0.3722
TSC2	auto	ok	88	42.86	25	0.2241	no	0.2546
PRKD1	auto	ok	16	66.67	25	0.2241	no	0.3225
SEC16A	auto	ok	17	80	50	0.2329	no	0.3752
AGO2	auto	ok	49	46.15	31.82	0.261	no	0.3209
STK11	auto	ok	97	14.29	30.43	0.2663	no	0.2657
NCOR1	auto	ok	80	0	35.29	0.2663	no	0.3513
KDR	auto	ok	28	25	33.33	0.2663	no	0.2071
SLX4	auto	ok	28	71.43	61.11	0.2663	no	0.383
IKBKE	auto	ok	27	50	36.36	0.2663	no	0.315
PARP1	auto	ok	22	20	40	0.2663	no	0.2573
CASP8	auto	ok	19	66.67	55.56	0.2663	no	0.263
ATM	auto	ok	72	12.5	32	0.2771	no	0.2111
CDKN2B	auto	ok	40	66.67	20	0.291	no	0.3909
CD74	auto	ok	74	68.42	89.39	0.2983	no	0.432
CCDC6	auto	ok	53	94.23	1.887	0.3042	no	0.4276
SUFU	auto	ok	17	66.67	25	0.3233	no	0.213
RPTOR	auto	ok	43	37.5	18.18	0.3408	no	0.2673
BCAS3	auto	ok	18	16.67	46.67	0.3408	no	0.2756
PAX8	auto	ok	43	42.86	80	0.3703	no	0.2854
INPPL1	auto	ok	39	57.14	53.85	0.3846	no	0.2332
ACOXL	auto	ok	8	14.29	50	0.4093	no	0.3113
KIF5B	auto	ok	95	97.7	96.88	0.4173	no	0.6266

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
ESR1	auto	ok	30	37.5	38.46	0.4361	no	0.2334
TOP1	auto	ok	31	0	50	0.4388	no	0.4077
WWOX	auto	ok	5	0	40	0.4388	no	0.4317
PRKAR1A	auto	ok	12	66.67	20	0.4534	no	0.2553
MAP3K1	auto	ok	42	33.33	30	0.4544	no	0.2556
CARM1	auto	ok	39	40	27.78	0.4544	no	0.2137
CREBBP	auto	ok	121	50	32.61	0.4566	no	0.2332
ERBB3	auto	ok	45	50	58.82	0.4885	no	0.2185
TFE3	auto	ok	45	87.5	80.56	0.4887	no	0.3534
NOTCH3	auto	ok	101	25	59.09	0.5044	no	0.1881
CSDE1	auto	ok	31	0	60	0.5044	no	0.4667
FH	auto	ok	29	16.67	13.33	0.5044	no	0.2761
TSC1	auto	ok	28	0	57.14	0.5044	no	0.4304
SND1	auto	ok	17	100	17.65	0.5044	no	0.3854
NRG1	auto	ok	16	100	86.67	0.5044	no	0.2663
FBXL20	auto	ok	8	100	28.57	0.5044	no	0.3592
FLT4	auto	ok	35	33.33	42.86	0.5191	no	0.2201
TRAF7	auto	ok	27	25	20	0.5454	no	0.2429
ABL1	auto	ok	21	0	25	0.5674	no	0.3956
KMT2C	auto	ok	134	14.29	53.57	0.5726	no	0.2041
DGKH	auto	ok	7	0	28.57	0.5726	no	0.4272
ELF3	auto	ok	47	50	55.56	0.578	no	0.2624
RAD51B	auto	ok	16	50	50	0.578	no	0.3065
ARID1B	auto	ok	72	0	58.33	0.5834	no	0.4781
AKT2	auto	ok	22	25	22.22	0.5834	no	0.2038
TTC28	auto	ok	15	28.57	63.64	0.5834	no	0.222
GOPC	auto	ok	6	75	50	0.5834	no	0.4432
FAT1	auto	ok	95	0	30	0.6126	no	0.4095
PLCG2	auto	ok	17	40	66.67	0.6233	no	0.2522
EXOC4	auto	ok	5	0	40	0.6238	no	0.5285
RAF1	auto	ok	23	37.5	50	0.6309	no	0.2143
ANKRD11	auto	ok	54	0	66.67	0.6571	no	0.4444
USP34	auto	ok	7	33.33	71.43	0.6964	no	0.4093
XPO1	auto	ok	31	12.5	41.67	0.6979	no	0.2217
MALT1	auto	ok	25	100	42.86	0.7036	no	0.2401
PIK3CB	auto	ok	20	0	60	0.7388	no	0.4483
CD79A	auto	ok	13	0	37.5	0.7388	no	0.4167

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
SDHA	auto	ok	12	66.67	60	0.7388	no	0.3269
FUBP1	auto	ok	29	50	66.67	0.7703	no	0.2052
ARID1A	auto	ok	148	44.44	37.93	0.7728	no	0.2075
CIC	auto	ok	68	0	47.37	0.7728	no	0.3363
KDM5A	auto	ok	44	50	55.56	0.7728	no	0.2317
BRCA1	auto	ok	66	25	52	0.788	no	0.1984
PPM1D	auto	ok	27	25	25	0.788	no	0.262
CACNA1A	auto	ok	7	33.33	50	0.788	no	0.2188
NELL1	auto	ok	5	50	80	0.788	no	0.3912
PRKAG2	auto	ok	5	25	50	0.788	no	0.2844
ARID5B	auto	ok	31	0	33.33	0.8094	no	0.4904
RB1	auto	ok	147	0	69.23	0.8116	no	0.3984
KMT2A	auto	ok	65	40	62.5	0.8116	no	0.1897
RAD21	auto	ok	38	100	55.56	0.8116	no	0.3412
PDGFRA	auto	ok	31	83.33	55.56	0.8116	no	0.2215
PICALM	auto	ok	5	0	60	0.8116	no	0.4194
CSF3R	auto	ok	17	66.67	66.67	0.8149	no	0.2944
KREMEN1	auto	ok	7	40	42.86	0.8275	no	0.2371
FOXA1	auto	ok	51	0	20	0.8377	no	0.5259
GLI1	auto	ok	51	57.14	60	0.8377	no	0.2434
BRIP1	auto	ok	46	0	23.53	0.8377	no	0.3595
NF2	auto	ok	46	20	11.11	0.8377	no	0.2264
NOTCH4	auto	ok	44	0	68.75	0.8377	no	0.3691
SMARCA4	auto	ok	143	36.84	41.67	0.8644	no	0.2563
DOT1L	auto	ok	85	55.56	36.36	0.8644	no	0.2478
RTEL1	auto	ok	57	50	37.5	0.8644	no	0.2172
PAK1	auto	ok	25	16.67	10	0.8644	no	0.2267
CCND1	auto	ok	15	0	57.14	0.8644	no	0.4286
KMT2D	auto	ok	128	25	48.57	0.8771	no	0.1893
WT1	auto	ok	119	93.94	4.808	0.8771	no	0.6209
EP300	auto	ok	98	14.29	31.82	0.8771	no	0.2053
MGA	auto	ok	65	33.33	46.67	0.8771	no	0.2372
RICTOR	auto	ok	34	60	55.56	0.8771	no	0.2787
INSR	auto	ok	31	50	50	0.8771	no	0.2214
PDGFRB	auto	ok	16	100	40	0.8771	no	0.2707
MYD88	auto	ok	8	66.67	33.33	0.8771	no	0.227
FANCA	auto	ok	49	30	35	0.8829	no	0.239

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
RARA	auto	ok	28	66.67	35.71	0.8829	no	0.2115
GRIN2A	auto	ok	19	0	20	0.8829	no	0.4012
LDLR	auto	ok	17	37.5	61.54	0.8829	no	0.3746
NUF2	auto	ok	14	100	11.11	0.8829	no	0.2888
NOS1AP	auto	ok	6	0	16.67	0.8829	no	0.4434
APC	auto	ok	110	0	45.45	0.8909	no	0.446
STAT3	auto	ok	51	100	11.11	0.8909	no	0.1969
AXL	auto	ok	37	75	25	0.8909	no	0.2142
SMARCB1	auto	ok	23	0	60	0.8909	no	0.5993
BLM	auto	ok	20	0	50	0.8909	no	0.4135
MTAP	auto	ok	8	33.33	25	0.8909	no	0.4351
NF1	auto	ok	247	27.78	47.73	0.9142	no	0.2742
AXIN1	auto	ok	29	100	66.67	0.9142	no	0.4256
PRKCI	auto	ok	22	66.67	50	0.9142	no	0.289
BAP1	auto	ok	67	42.86	33.33	0.9481	no	0.2515
ERBB4	auto	ok	42	100	40	0.9481	no	0.2312
KDM5C	auto	ok	33	100	66.67	0.9481	no	0.2085
TP53BP1	auto	ok	31	66.67	66.67	0.9481	no	0.2741
MST1R	auto	ok	13	33.33	40	0.9481	no	0.2528
PTK2	auto	ok	8	66.67	50	0.9481	no	0.2708
IGF2	auto	ok	7	100	75	0.9481	no	0.2614
IRS2	auto	ok	26	50	77.78	0.9517	no	0.2554
BRD4	auto	ok	86	11.11	39.29	0.9633	no	0.1913
BRCA2	auto	ok	79	14.29	63.64	0.9633	no	0.214
MTOR	auto	ok	49	33.33	30	0.9633	no	0.2159
TBX3	auto	ok	36	0	60	0.9633	no	0.4012
PTPRT	auto	ok	33	100	60	0.9633	no	0.221
PDE4D	auto	ok	8	25	66.67	0.9633	no	0.3623
PTPRS	auto	ok	44	25	38.46	0.9831	no	0.1952
SMARCD1	auto	ok	21	33.33	57.14	0.9831	no	0.2672
PTEN	auto	ok	86	50	25	1	no	0.2667
PBRM1	auto	ok	85	16.67	50	1	no	0.2127
ZFXH3	auto	ok	85	0	16.67	1	no	0.5238
DNMT1	auto	ok	80	45.45	20	1	no	0.2288
ARID2	auto	ok	72	0	23.08	1	no	0.4156
CTNNB1	auto	ok	72	60	44.44	1	no	0.216
KDM6A	auto	ok	71	50	100	1	no	0.09149

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ATRX	auto	ok	64	50	42.86	1	no	0.281
SMAD4	auto	ok	62	0	16.67	1	no	0.4259
SPEN	auto	ok	61	0	27.78	1	no	0.357
CDH1	auto	ok	58	50	85.71	1	no	0.2045
PTPRD	auto	ok	57	25	42.86	1	no	0.2174
PIK3R1	auto	ok	55	50	33.33	1	no	0.1538
MDC1	auto	ok	52	50	57.14	1	no	0.2447
ATR	auto	ok	48	0	33.33	1	no	0.3882
SETD2	auto	ok	46	0	50	1	no	0.3671
NSD1	auto	ok	45	50	36.36	1	no	0.2471
PIK3C2G	auto	ok	43	50	33.33	1	no	0.2406
B2M	auto	ok	42	0	0	1	no	0.6666
DNMT3B	auto	ok	40	0	16.67	1	no	0.3882
KEAP1	auto	ok	40	0	33.33	1	no	0.3785
NFE2L2	auto	ok	40	0	20	1	no	0.5001
EZH2	auto	ok	39	40	7.143	1	no	0.2783
FOXP1	auto	ok	39	50	80	1	no	0.2861
NBN	auto	ok	38	0	33.33	1	no	0.4434
PREX2	auto	ok	37	0	50	1	no	0.4466
BCOR	auto	ok	36	0	57.14	1	no	0.3882
DROSHA	auto	ok	36	50	14.29	1	no	0.3074
IGF1R	auto	ok	36	33.33	25	1	no	0.2019
NCOA3	auto	ok	33	50	75	1	no	0.2059
RUNX1	auto	ok	33	50	33.33	1	no	0.2765
BCL2L11	auto	ok	32	23.08	21.43	1	no	0.3364
JAK3	auto	ok	32	0	37.5	1	no	0.392
PIK3R2	auto	ok	32	20	40	1	no	0.2632
RPS6KB2	auto	ok	32	50	0	1	no	0.2761
ERCC2	auto	ok	31	50	75	1	no	0.2651
ETV1	auto	ok	30	68.75	0	1	no	0.3978
GNAS	auto	ok	30	40	0	1	no	0.2779
CALR	auto	ok	29	80	0	1	no	0.2806
CBL	auto	ok	29	50	50	1	no	0.3399
JAK1	auto	ok	29	50	0	1	no	0.2064
NTRK2	auto	ok	29	0	40	1	no	0.4405
AXIN2	auto	ok	28	0	75	1	no	0.4466
TEK	auto	ok	28	0	55.56	1	no	0.3709

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TET1	auto	ok	28	100	28.57	1	no	0.2135
MAP2K4	auto	ok	27	50	0	1	no	0.3107
SMAD3	auto	ok	27	33.33	55.56	1	no	0.2404
SOX9	auto	ok	26	0	25	1	no	0.4441
UPF1	auto	ok	26	100	60	1	no	0.232
CARD11	auto	ok	25	0	28.57	1	no	0.4304
FLT1	auto	ok	25	33.33	33.33	1	no	0.1589
ASXL2	auto	ok	24	66.67	57.14	1	no	0.2761
DIS3	auto	ok	24	50	16.67	1	no	0.3189
TP63	auto	ok	24	66.67	14.29	1	no	0.2747
AKT1	auto	ok	23	0	42.86	1	no	0.5238
MAP3K13	auto	ok	23	0	20	1	no	0.4385
MDM2	auto	ok	23	66.67	47.06	1	no	0.2584
MED12	auto	ok	23	0	33.33	1	no	0.4169
STAT5A	auto	ok	23	33.33	0	1	no	0.2083
TCF7L2	auto	ok	23	0	50	1	no	0.5
EIF4A2	auto	ok	22	0	33.33	1	no	0.4012
PIK3CD	auto	ok	22	33.33	50	1	no	0.2162
PLK2	auto	ok	22	50	57.14	1	no	0.3264
PPP2R1A	auto	ok	22	75	33.33	1	no	0.2948
RAD54L	auto	ok	22	0	60	1	no	0.4194
FLT3	auto	ok	21	0	20	1	no	0.4441
MAP2K2	auto	ok	21	40	14.29	1	no	0.2448
FLCN	auto	ok	20	71.43	44.44	1	no	0.2682
MSH6	auto	ok	20	0	80	1	no	0.4012
PIK3R3	auto	ok	20	100	0	1	no	0.1419
SMYD3	auto	ok	20	50	33.33	1	no	0.1125
SRC	auto	ok	20	100	0	1	no	0.2477
CHEK2	auto	ok	19	25	16.67	1	no	0.2718
CSF1R	auto	ok	19	0	33.33	1	no	0.3709
LATS1	auto	ok	19	50	33.33	1	no	0.2232
PTCH1	auto	ok	19	0	20	1	no	0.4317
YAP1	auto	ok	19	50	20	1	no	0.2861
FGFR4	auto	ok	18	50	44.44	1	no	0.2087
PGR	auto	ok	18	33.33	57.14	1	no	0.2665
VTCN1	auto	ok	18	50	60	1	no	0.288
EGFL7	auto	ok	17	66.67	25	1	no	0.4086

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ERCC5	auto	ok	17	0	60	1	no	0.5
SDC4	auto	ok	17	100	0	1	no	0.4731
TAP1	auto	ok	17	0	50	1	no	0.4286
TGFBR1	auto	ok	17	100	75	1	no	0.2714
CTCF	auto	ok	16	0	50	1	no	0.5
CUL1	auto	ok	16	100	0	1	no	0.3837
RECQL	auto	ok	16	0	0	1	no	0.4713
SMO	auto	ok	16	0	12.5	1	no	0.4032
SUZ12	auto	ok	16	0	20	1	no	0.4194
CCNE1	auto	ok	15	50	0	1	no	0.2861
MSH3	auto	ok	15	0	50	1	no	0.5
TNFRSF14	auto	ok	15	100	0	1	no	0.2625
CDK4	auto	ok	14	66.67	0	1	no	0.1183
E2F3	auto	ok	14	0	20	1	no	0.4194
FOXO1	auto	ok	14	100	0	1	no	0.3721
MPL	auto	ok	14	0	83.33	1	no	0.4665
NUP93	auto	ok	14	0	25	1	no	0.4194
SESN3	auto	ok	14	50	0	1	no	0.1214
SF3B1	auto	ok	14	33.33	40	1	no	0.2872
EZR	auto	ok	13	66.67	0	1	no	0.4151
NPM1	auto	ok	13	100	50	1	no	0.3172
TRAPPC9	auto	ok	13	42.86	0	1	no	0.4889
YES1	auto	ok	13	50	16.67	1	no	0.2505
AKT3	auto	ok	12	100	50	1	no	0.2617
CEBPA	auto	ok	12	0	0	1	no	0.4444
DUSP4	auto	ok	12	0	33.33	1	no	0.4135
FYN	auto	ok	12	100	66.67	1	no	0.1422
IDH1	auto	ok	12	100	0	1	no	0.2625
PAX5	auto	ok	12	100	66.67	1	no	0.186
PIK3C3	auto	ok	12	100	100	1	no	0.3772
PIK3CA	auto	ok	12	0	80	1	no	0.5222
ERCC4	auto	ok	11	33.33	50	1	no	0.2872
FGF19	auto	ok	11	0	0	1	no	0.4888
GSK3B	auto	ok	11	100	0	1	no	0.2625
MKRN1	auto	ok	11	77.78	0	1	no	0.3815
MLH1	auto	ok	11	100	50	1	no	0.2625
PHF7	auto	ok	11	100	0	1	no	0.2625

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
RAD51D	auto	ok	11	100	0	1	no	0.2625
RELA	auto	ok	11	66.67	100	1	no	0.426
SOS1	auto	ok	11	50	0	1	no	0.1471
TACC2	auto	ok	11	100	100	1	no	0.5485
BMPR1A	auto	ok	10	100	0	1	no	0.3772
CDK8	auto	ok	10	100	0	1	no	0.1472
CUL3	auto	ok	10	50	0	1	no	0.2599
EPAS1	auto	ok	10	0	20	1	no	0.4194
IKZF3	auto	ok	10	50	62.5	1	no	0.2719
MDM4	auto	ok	10	50	50	1	no	0.3137
RXRA	auto	ok	10	50	66.67	1	no	0.1419
TRIM24	auto	ok	10	100	0	1	no	0.3932
ARAF	auto	ok	9	100	50	1	no	0.2024
LRP1	auto	ok	9	0	20	1	no	0.4347
NTHL1	auto	ok	9	50	66.67	1	no	0.1961
PPARG	auto	ok	9	100	50	1	no	0.2072
RAD51	auto	ok	9	100	66.67	1	no	0.2773
SESN1	auto	ok	9	0	0	1	no	0.4194
STK40	auto	ok	9	33.33	0	1	no	0.1953
ETV4	auto	ok	8	100	16.67	1	no	0.3893
HNF1A	auto	ok	8	0	25	1	no	0.4441
IKZF1	auto	ok	8	100	0	1	no	0.3772
IRF4	auto	ok	8	0	25	1	no	0.4667
KRAS	auto	ok	8	100	0	1	no	0.1856
MAPK1	auto	ok	8	50	0	1	no	0.2599
MCL1	auto	ok	8	0	60	1	no	0.4538
MSI1	auto	ok	8	25	40	1	no	0.2176
SLC25A21	auto	ok	8	0	14.29	1	no	0.4466
STRN	auto	ok	8	100	0	1	no	0.5381
CDK6	auto	ok	7	50	0	1	no	0.2599
CREB1	auto	ok	7	80	71.43	1	no	0.3515
CRKL	auto	ok	7	0	50	1	no	0.6442
DCUN1D1	auto	ok	7	100	0	1	no	0.1596
DNM2	auto	ok	7	40	42.86	1	no	0.228
GAB2	auto	ok	7	50	20	1	no	0.2861
ST7	auto	ok	7	33.33	0	1	no	0.437
TACC1	auto	ok	7	100	100	1	no	0.2625

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
CAMTA1	auto	ok	6	100	0	1	no	0.1241
CDKN2C	auto	ok	6	100	66.67	1	no	0.1565
CPM	auto	ok	6	50	0	1	no	0.1385
CTNNA3	auto	ok	6	75	0	1	no	0.2525
DTNB	auto	ok	6	100	0	1	no	0.3383
EIF3H	auto	ok	6	100	0	1	no	0.529
EPHX3	auto	ok	6	100	0	1	no	0.2625
SMARCA2	auto	ok	6	50	66.67	1	no	0.1424
SOX2	auto	ok	6	0	75	1	no	0.5009
TANC2	auto	ok	6	100	50	1	no	0.2271
TFG	auto	ok	6	100	100	1	no	0.3246
ZNRF3	auto	ok	6	50	66.67	1	no	0.4336
ATE1	auto	ok	5	50	0	1	no	0.3752
BCL2L1	auto	ok	5	50	50	1	no	0.2093
CNTNAP2	auto	ok	5	66.67	0	1	no	0.267
CREM	auto	ok	5	20	100	1	no	0.2602
CTTNBP2	auto	ok	5	0	20	1	no	0.5285
EIF4E	auto	ok	5	50	0	1	no	0.2599
FBXL7	auto	ok	5	100	75	1	no	0.4249
GPATCH8	auto	ok	5	50	50	1	no	0.3093
HCN1	auto	ok	5	100	0	1	no	0.3772
LRP1B	auto	ok	5	100	100	1	no	0.2008
MAD1L1	auto	ok	5	40	0	1	no	0.2382
RAC2	auto	ok	5	50	0	1	no	0.2599
SBNO2	auto	ok	5	0	75	1	no	0.4466
SDHAF2	auto	ok	5	100	0	1	no	0.4933
SFPQ	auto	ok	5	50	75	1	no	0.4008
SHANK2	auto	ok	5	0	25	1	no	0.4466
SMARCE1	auto	ok	5	100	0	1	no	0.4933
SMURF2	auto	ok	5	100	75	1	no	0.2569
SPTBN1	auto	ok	5	100	0	1	no	0.3932
THSD4	auto	ok	5	50	25	1	no	0.2439
TPR	auto	ok	5	100	0	1	no	0.5142
CDKN2A	auto	ok	185	14.29	0			0.2857
KMT2B	unresolved	failed	68					
NSD3	auto	ok	58	0	0			0.1429
RECQL4	unresolved	failed	54					

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
PRKN	auto	ok	40	0	0			0.25
NSD2	auto	ok	37	0	0			0.3333
MAP3K14	unresolved	failed	36					
ASXL1	auto	ok	35	0	100			0.2222
TCF3	unresolved	failed	35					
RASA1	auto	ok	32	0	66.67			0.5
STAG2	auto	ok	31	0	0			0.5
AR	auto	ok	30	0	50			0.5
CDC73	auto	ok	28	0	50			0.5
JAK2	auto	ok	27	0	33.33			0.3016
CDKN1B	auto	ok	24	0	0			0.3016
MSH2	auto	ok	23	0	0			0.2731
NFKBIA	auto	ok	23	0	100			1
MEN1	auto	ok	22	0	0			0.3016
MYC	auto	ok	22	0	0			0.3333
TET2	auto	ok	21	0	66.67			0.5516
ERF	unresolved	failed	20					
KIT	auto	ok	20	0	0			0.4266
CDKN2B-AS1	unresolved	failed	19					
EPHA3	auto	ok	19	0	0			0.4266
NEGR1	auto	ok	19	0	66.67			0.3333
PALB2	auto	ok	19	0	0			0.2
SEPTIN14	unresolved	failed	19					
BBC3	auto	ok	18	100	0			0.375
GATA3	auto	ok	18	0	0			0.5
AMER1	auto	ok	17	0	50			0.3333
COP1	auto	ok	17	0	0			0.5
DDR2	auto	ok	17	0	0			0.2016
EPHA5	auto	ok	17	0	33.33			0.5516
INPP4B	auto	ok	17	100	0			1
RAD50	auto	ok	17	0	33.33			0.3016
BCL6	auto	ok	16	0	66.67			0.3016
DICER1	auto	ok	16	0	0			0.3856
ERRF1	auto	ok	16	0	0			0.6667
LATS2	auto	ok	16	0	0			0.8016
MITF	auto	ok	16	0	100			1
MSI2	auto	ok	16	0	50			0.4266

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
NCOA4	auto	ok	16	0	0			1
RAD52	auto	ok	16	0	100			0.8016
TGFBR2	auto	ok	16	0	100			0.5
TRAP1	unresolved	failed	16					
CREB3L1	unresolved	failed	15					
INPP4A	auto	ok	15	33.33	0			0.2857
SRP19	auto	ok	15	0	0			1
GPS2	auto	ok	14	0	0			0.5
HGF	auto	ok	14	0	66.67			0.4606
LINC00114	unresolved	failed	14					
RIT1	auto	ok	14	0	0			0.3596
TAP2	auto	ok	14	0	0			0.5
TRAF2	unresolved	failed	14					
CYLD	auto	ok	13	0	0			1
DAXX	auto	ok	13	0	0			1
MAP2K1	auto	ok	13	0	66.67			0.25
MEF2B	unresolved	failed	13					
MRE11	auto	ok	13	0	100			0.2
PAK5	auto	ok	13	0	0			0.5
RAD51C	auto	ok	13	0	0			0.1766
SPOP	auto	ok	13	0	50			0.4266
BIRC3	auto	ok	12	0	66.67			0.3016
BTK	auto	ok	12	0	0			1
CBFB	auto	ok	12	0	0			1
EED	auto	ok	12	0	0			0.4266
IRS1	auto	ok	12	0	0			0.5
RAB35	auto	ok	12	0	0			0.3356
SMAD2	auto	ok	12	0	0			0.8016
VEGFA	auto	ok	12	0	0			0.4606
ALOX12B	auto	ok	11	0	0			0.4266
BABAM1	auto	ok	11	0	0			0.5
BCL2	auto	ok	11	0	0			1
EPHA7	auto	ok	11	0	50			0.4266
FANCC	auto	ok	11	0	0			0.6096
FBXW7	auto	ok	11	0	0			0.3016
IDH2	auto	ok	11	0	0			0.4266
SRSF2	auto	ok	11	0	100			1

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
TNFAIP3	auto	ok	11	0	0			0.3016
AURKA	auto	ok	10	0	33.33			0.5516
CCND2	auto	ok	10	0	100			0.3516
EPHB1	auto	ok	10	0	100			0.5
H3C2	unresolved	failed	10					
MAPK3	auto	ok	10	0	0			0.8016
PIK3CG	auto	ok	10	0	0			
PMS1	auto	ok	10	0	0			
PTPN11	auto	ok	10	0	0			0.3333
SH2B3	auto	ok	10	0	33.33			0.25
SOX17	auto	ok	10	0	0			0.5856
SYK	auto	ok	10	0	0			1
CHEK1	auto	ok	9	0	0			0.4266
GATA1	auto	ok	9	0	0			1
IFNGR1	auto	ok	9	0	33.33			0.3016
MAPKAP1	auto	ok	9	0	100			0.8016
PPP6C	auto	ok	9	0	0			1
PRDM14	auto	ok	9	0	0			
REL	auto	ok	9	0	0			
SHQ1	auto	ok	9	0	100			0.5
AKAP8	auto	ok	8	0	0			1
AURKB	auto	ok	8	0	33.33			0.3333
BARD1	auto	ok	8	0	0			
BCL2L14	auto	ok	8	66.67	0			0.75
CCND3	auto	ok	8	0	0			0.8356
CD79B	auto	ok	8	0	33.33			0.3016
ERCC3	auto	ok	8	0	33.33			0.3016
FGF3	auto	ok	8	0	0			0.3333
GNA11	auto	ok	8	0	0			1
ICOSLG	auto	ok	8	0	0			
LYN	auto	ok	8	0	0			0.2391
NBPF20	unresolved	failed	8					
PDCD1	auto	ok	8	0	0			1
RRAS	auto	ok	8	0	33.33			0.3333
XRCC2	auto	ok	8	0	0			1
ACP3	auto	ok	7	0	0			0.4286
AP1B1	auto	ok	7	0	0			1

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
ASIC2	auto	ok	7	0	0			0.3333
ATP1B2	auto	ok	7	0	0			1
DCBLD1	auto	ok	7	0	0			
EPCAM	auto	ok	7	0	0			0.5
H3C4	auto	ok	7	0	0			
INHA	auto	ok	7	0	0			1
JUN	auto	ok	7	0	0			
KLF4	auto	ok	7	0	0			
NKX3-1	auto	ok	7	0	0			1
PDPK1	auto	ok	7	0	100			1
RAB11FIP4	unresolved	failed	7					
RYBP	auto	ok	7	0	0			
SDHC	auto	ok	7	0	66.67			0.3016
SHOC2	auto	ok	7	0	100			1
TENT5C	auto	ok	7	0	0			
AGAP3	unresolved	failed	6					
CD274	auto	ok	6	0	0			1
CD276	auto	ok	6	0	0			
CDH3	unresolved	failed	6					
FGF4	auto	ok	6	0	50			0.5
GALNT11	auto	ok	6	0	66.67			1
GRB7	auto	ok	6	0	33.33			0.8
H1-2	auto	ok	6	0	0			1
H3-3B	auto	ok	6	0	0			0.5
H3C6	unresolved	failed	6					
HOXB13	auto	ok	6	0	0			0.5

Showing the first 500 of 544 scanned genes.

Honorable mentions: highly ranked non-FDR-significant genes worth human review

Did NOT survive genome-wide FDR correction -- ranked by raw Fisher p-value among non-significant genes. Not a claim of significance.

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret. %
1	RET	0.0004197	0.1066	194	85.38	92.27
2	FGFR2	0.0004247	0.1066	136	93.1	85.82

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret. %
3	ALK	0.001917	0.1675	272	97.41	96.31
4	EGFR	0.01145	0.1917	55	70	74.55
5	BRAF	0.01337	0.1675	179	92.64	91.57
6	FGFR3	0.01948	0.1675	152	97.37	94.04
7	CDKN2B	0.02222	0.291	12	66.67	20
8	IKBKE	0.02424	0.2663	12	50	36.36
9	TMPRSS2	0.0292	0.1675	867	33.33	6.243
10	NTRK1	0.04826	0.1675	78	72.73	84.21
11	INPPL1	0.04895	0.3846	14	57.14	53.85
12	CDK12	0.0677	0.1675	43	50	39.02
13	NAB2	0.1126	0.1675	72	71.43	69.01
14	KDM5A	0.119	0.7728	10	50	55.56
15	FH	0.1333	0.5044	16	16.67	13.33

Figures

manhattan

